

Inheritance Program

Program Code:

```
class A {
    void smtg() {
        System.out.println("This is Class A");
    }
}

class B extends A {
    void smtg2() {
        System.out.println("This is Class B");
    }
}

class C extends A {
    void smtg3() {
        System.out.println("This is Class C");
    }
}

class D extends B {
    void smtg4() {
        System.out.println("This is Class D");
    }
}

public class Main {
    public static void main(String[] args) {

        D obj = new D();

        obj.smtg();
        obj.smtg2();
        obj.smtg4();

        C obj2 = new C();

        obj2.smtg();
        obj2.smtg3();
    }
}
```

Explanation:

This program demonstrates inheritance between multiple classes. Class B and class C extend class A, and class D extends class B. An object of class D can access the methods of classes A, B, and D. Another object of class C can access the methods of classes A and C. The program calls these methods and displays the corresponding messages.

Compilation and Execution Commands:

Save the file as: Main.java

Compile:
javac Main.java

Run:
java Main

Output Screenshot:

```
This is Class A
```

```
This is Class B
```

```
This is Class D
```

```
This is Class A
```

```
This is Class C
```

Verification:

The output confirms that the object of class D accesses the inherited method from class A, the method of class B, and its own method. The object of class C accesses the inherited method from class A and its own method successfully.

Result:

Thus, the Java program was successfully executed and the inheritance relationships between classes A, B, C, and D were verified successfully.